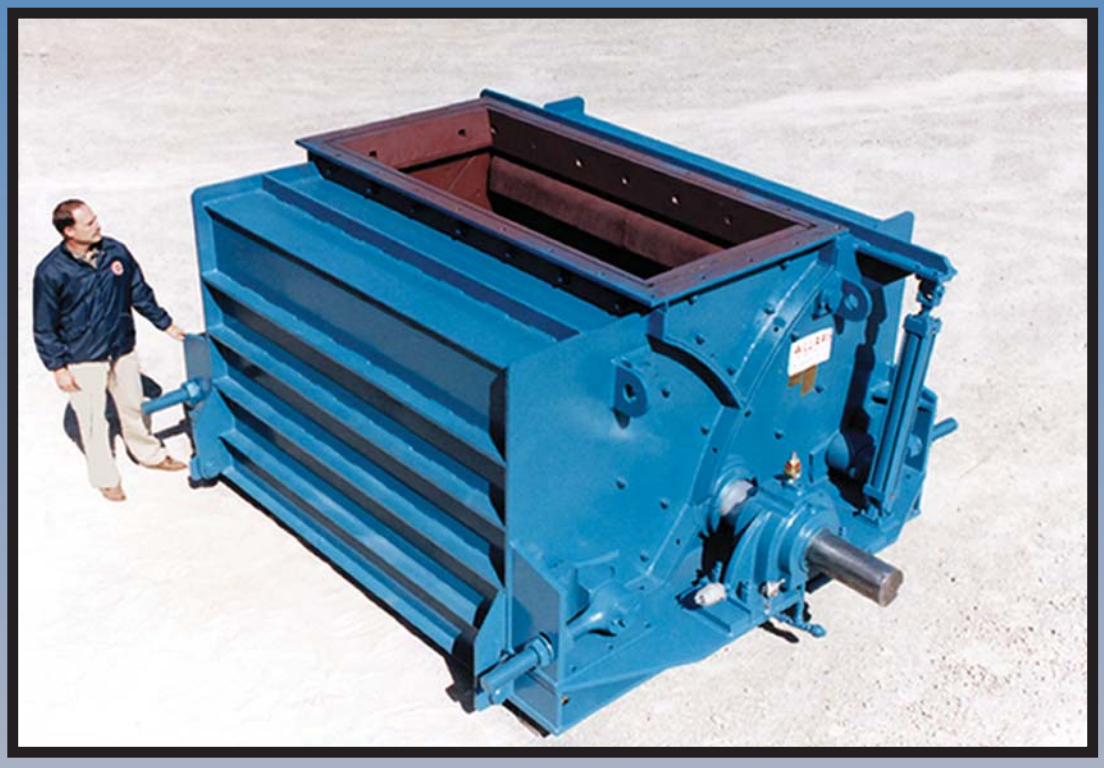
REVERSIBLE IMPACTORS



HIGH CAPACITY SIZE REDUCTION OF SOLIDS BY MID-AIR IMPACT

WILLIAMS

Patent Crusher & Pulverizer Co., Inc.

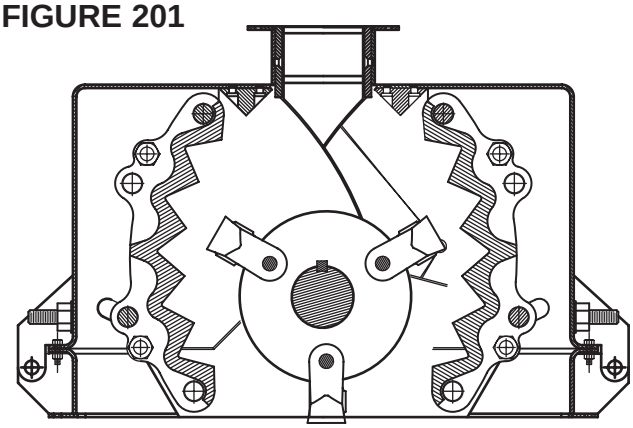
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BULLETIN 855R

Williams Reversible Impactors

- **High Capacity Size Reduction of Solids by Mid-Air Impact**
- **Well Formed Cubical Product --- Minimum Flats & Silvers**
- **Large Reduction Ratio --- Up to 35 to 1 In One Stage**
- **Unrestricted Discharge Opening**

FIGURE 201



Typical of the construction of a Williams Reversible Impactor, hammers are free to pivot back to accommodate uncrushables. Impact sections are reversible for long life and adjustable to maintain optimum crushing clearances. The cover sections (one or both) provide full access to the interior without need for disturbing the feed chute.

Williams Reversible Impactors are among the most advanced of the secondary or tertiary crushers for the reduction of stone, coal, ore, clay, shale, refractories, ceramics, scrap glass, chemicals, fertilizers, and other materials.

Impact crushing typically fractures material across the line of cleavage rather than crushing only with the strata. Consequently, the product formed in an Impactor contains a high percentage of cube-shaped material. When compared to the product from machines that crush by attrition or compression, the product from an Impactor contains far less material with flats and silvers.

Size reduction ratios as high as 35 to 1 are frequently obtained with an Impactor. Often, one Impactor can do the job of two crushing machines, thereby saving the cost of additional motors, conveyors, and surge piles.

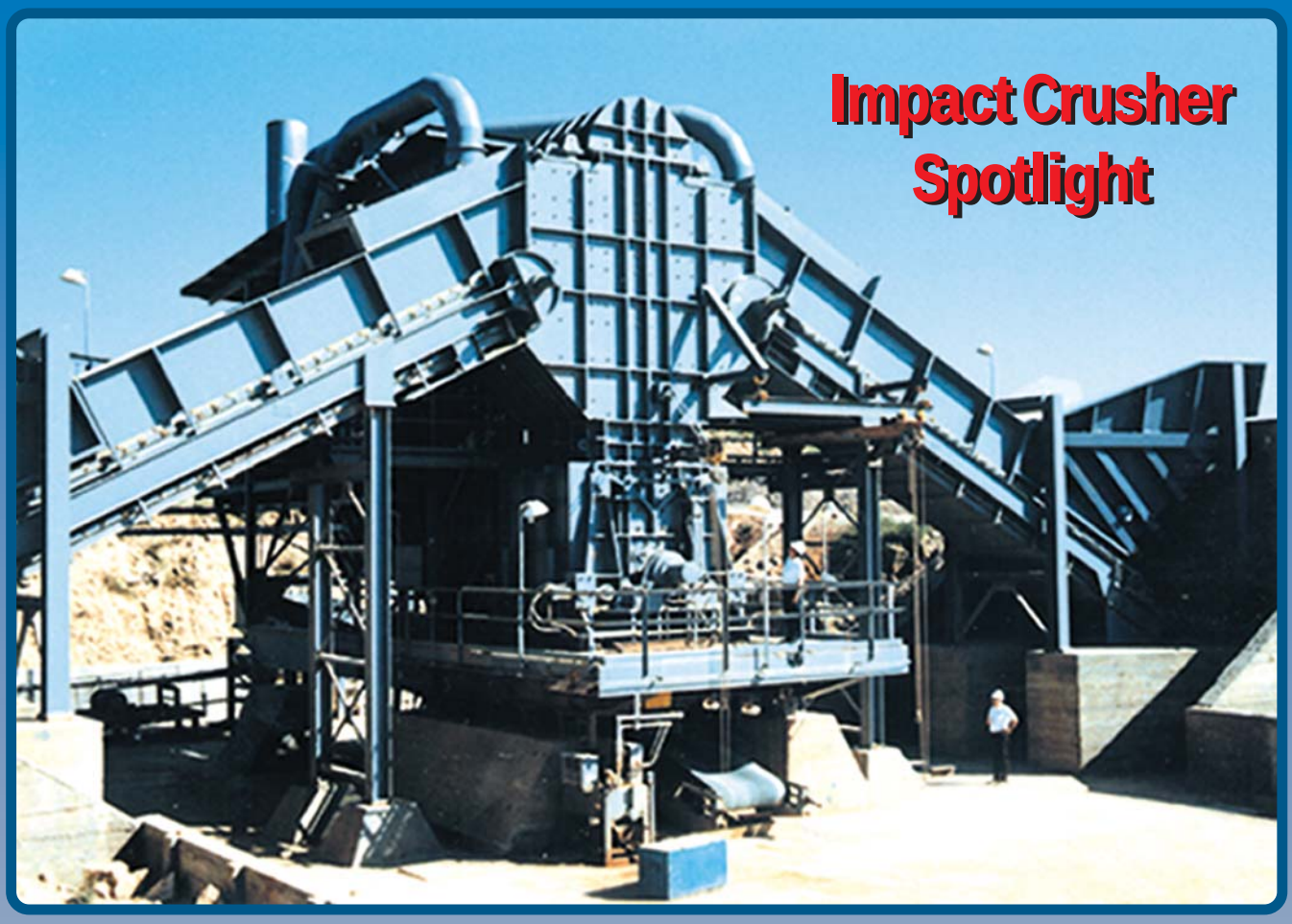
The Impactor, with its large unrestricted discharge opening, can frequently crush wet or sticky materials that would plug other crusher types.

With an Impactor crushing system, maintenance and operating costs can be drastically reduced. There are few close tolerances between the crushing surfaces, and no cage bars to maintain. Horsepower per ton is generally quite low.

Push button reversibility of the rotor permits the use of both sides of the hammer without resorting to costly manual hammer turning. This assures maximum utilization of crushing surfaces.

WILLIAMS ***. . . THE LEADERS IN SIZE REDUCTION***

Impact Crusher Spotlight



Shown above is a 2000 HP Williams Impact Crusher at a large cement plant in Morocco. The #780 Reversible Impactor is used as a primary crusher to reduce 40” limestone to a minus 1” product at 900 - 1000 Tons per Hour in an open circuit.



Shovel-loaded limestone and shale are fed to the Impact Crusher by two Williams Heavy-Duty, Piano-Hinge steel conveyors. End product travels three miles to cement plant on movable conveyor.



#780 Impact Crusher with reinforced hopper and hydraulic cover opening system for easy maintenance.

FEATURES



FIGURE 401



FIGURE 402



FIGURE 403

Impact Blocks

Williams Reversible Impactors utilize one piece impact blocks that have proven to be far superior to old design, multiple, bolt-on breaker blocks.

The Williams Impact Block is designed to be rotated top for bottom to assure maximum utilization of the part. Positive positioning of the Impact Block by means of heavy jack bolts, or with an optional hydraulic system, permits maintenance of the optimum internal crushing clearances, thereby assuring continuing conformance with product specifications.

A most important feature for the user – with Williams one piece Impact Block, typical problems encountered in changing multiple, bolt-on breaker blocks are eliminated. No longer does the user contend with inaccessible old-down bolts that have seized or loosened, or with rough multiple replacement castings that don't match, won't fit, and take longer to install. *SEE FIGURE 401.*

Rotors

Rotors are available in diameters up to 72" and in lengths up to 120" for the largest crushing requirements. The hammers are alloy steel and heat-treated for resistance to abrasion. Shafts are of forged heat-treated steel carefully machined to size and fitted with keyways for the discs and drive member. The heavy construction of the Williams rotor provides sufficient moment of inertia so that an internal flywheel effect is created. An external flywheel is not required. For quiet operation, rotors are statically and dynamically balanced prior to shipment. *SEE FIGURE 402.*



FIGURE 404

Hammers

Hammer design and metallurgy provide the optimum weight distribution and wear resistance for Impact Crushing. Hammers are of the “T” or yoke design suspended from heat-treated hammer bolts. Hammers are free to pivot back in the event tramp iron enters the Impactor. *SEE FIGURE 403.*

Bearings

Heavy-duty, self-aligning, double row, anti-friction bearings are provided as standard. Depending on the size and speed of the rotor, grease lubrication, static oil lubrication, or circulating oil lubrication is provided. Large shafts are designed for hydraulic fitting and removal of the inner bearing race. *SEE FIGURE 501.*



FIGURE 501

Power Train

Williams Reversible Impactors are suitable for V-Belt drive, direct connection through a flexible coupling, drive through a gear reducer, or dual drive from each end of the shaft. Motor starters should be of the reversing type.

Full Access Covers

Full Access Covers provide quick, complete access to the Impactor interior. The covers can be opened without disturbing the feed chute or overhead feeder mechanisms. Both sides of the Impactor frame can be made removable, hinged, or with hydraulic openers added. *SEE FIGURES 404 and 503.*



FIGURE 502: Jack bolts provide for adjustment of the impact sections. Shims maintain proper adjustment and transmit crushing forces to the rugged steel plate frame.





FIGURE 503: Hydraulic openers have been added to this Reversible Impactor to provide full access to Impactor interior.

Williams Reversible Impactors in Closed Circuit Systems

Frequently a Reversible Impactor works to best advantage in a closed circuit crushing system. Such a system will permit:

- **High Impactor Capacity**
- **Large Crushing Clearances for Low Wear and Low Power Requirement**
- **Positive Control of Product Top Size**

In closed circuit systems, a screen following the Impactor passes only that material which has been reduced below the specified top size. Material rejected by the screen is returned to the Impactor for additional crushing.

Tailings from the screen constitute a circulating load. For the purpose of selecting an Impactor of adequate capacity, the circulating load must be included with the uncrushed feed. Typically, the circulating load will amount to 20% or more of the overall system capacity.



Typical Impactor Product Characteristics

The Williams Reversible Impactor will produce a product that is cubical in shape and contains a minimum of slivers. Control of the product size gradation is possible by varying rotor speed and the clearance between the hammer circle and the Impact Block.

Close internal clearance and high rotor tip speed will tend to reduce the size of the crushed material. This is illustrated in the accompanying Table #1.

This table can be used to estimate the gross capacity of the Impactor required to produce a specified tonnage of material less than a given top size. The results are based on an average of several medium limestones. To obtain Gross Capacity, divide by the appropriate percentage.

TABLE 1

ONE PASS ANALYSIS -- TYPICAL PRODUCT SIZE GRADATION

Screen Opening (inches)	Cumulative Percent Passing at Indicated Rotor Tip Speed		
	11,000 ft./min.	8850 ft./min.	7250 ft./min.
2		99	92
1 1/2	100	98	85
1	95	85	75
3/4	90	75	60
5/8	85	70	55
1/2	80	60	50
3/8	68	50	40
1/4	55	40	30
3/16	50	35	25
1/8	40	30	20

WILLIAMS

. . . THE LEADERS IN SIZE REDUCTION . . .

Table 2

Dimensions are furnished as guides and are subject to change. Use certified prints for final engineering layout.

Allow ample room above Impactor for Feeder and Feed Chute. Allow room for removal of Hammer Bolts and for removal or pivoting of Access Cover or Covers. Common Motor Base is not included. Refer to factory for minimum distance for servicing.

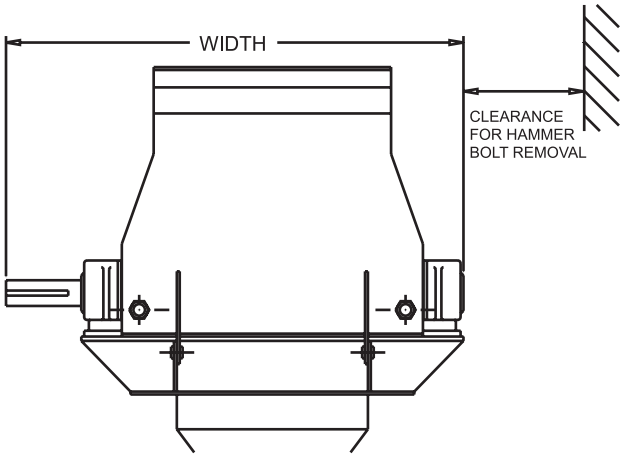
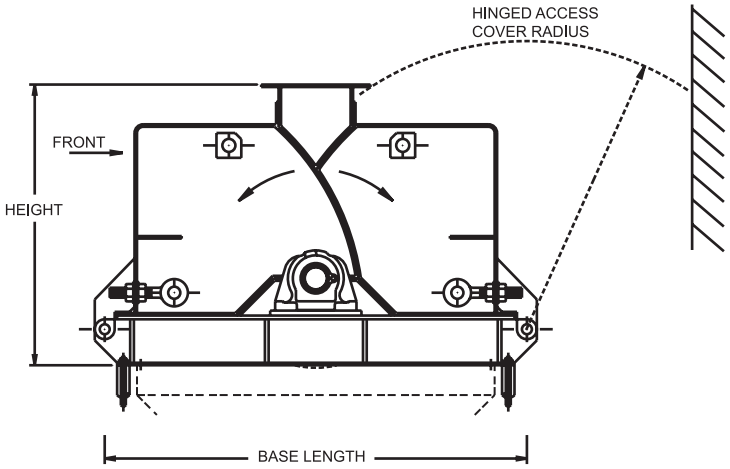


Table 3

Capacities shown in the chart should serve as a guide and are basedon a uniform feed rate of the maximum allowable feed size evenly distributed across the feed opening. Material is medium Limestone with a work index of 10 crushed to 80% passing 5/8". Williams Engineers are prepared to recommend the proper machine for other materials and reduction ratios.



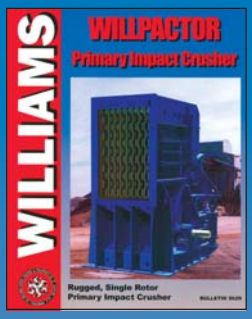
DIMENSIONS TABLE 2

SIZE	BASE LENGTH (inches)	WIDTH (inches)	HEIGHT (inches)	FEED OPENING (inches)	DISCHARGE OPENING (inches)	CLEARANCE for COVERS (inches)	CLEAR for HAMMER BOLT (inches)	WEIGHT (Pounds)
111	31.5	51	23.5	6 x 11	18.75 x 11	N/A	0	1,256
215	62.5	60	38.5	10 x 16	57.5 x 18	N/A	0	5,400
220	62.5	67	38.5	10 x 21	57.5 x 23	N/A	5	5,700
230	62.5	77	38.5	10 x 31	57.5 x 33	N/A	15	6,600
240	62.5	87	38.5	10 x 41	57.5 x 43	N/A	25	7,800
320	80	76	56.5	12 x 20.5	74 x 23	58	0	11,850
330	80	88	56.5	12 x 30.5	74 x 33	58	10	13,800
340	80	100	56.5	12 x 40.5	74 x 43	58	20	15,500
350	80	112	56.5	12 x 50.5	74 x 53	58	30	17,800
360	80	122	56.5	12 x 60.5	74 x 63	58	40	21,200
370	80	133	56.5	12 x 70.5	74 x 73	58	54	23,600
380	80	147	56.5	12 x 80.5	74 x 83	58	58	27,000
430	104	99	69	17 x 30	98 x 35	69	15	33,400
445	104	114	69	17 x 45	98 x 50	69	30	37,400
460	104	130	69	17 x 60	98 x 65	69	42	43,100
475	104	141	69	17 x 75	98 x 80	69	57	50,000
490	104	156	69	17 x 90	98 x 95	69	72	57,000
540	118	112	79	22 x 40	111 x 45	78	20	47,000
560	118	132	79	22 x 60	111 x 65	78	40	55,700
580	118	152	79	22 x 80	111 x 85	78	60	66,500
5100	118	174	79	22 x 100	111 x 105	78	80	80,500
5120	118	194	79	22 x 120	111 x 125	78	100	95,000
640	120	114	89	27 x 40	113 x 46	85	20	58,500
660	120	134	89	27 x 60	113 x 66	85	40	70,000
680	120	154	89	27 x 80	113 x 86	85	60	82,000
6100	120	175	89	27 x 100	113 x 106	85	80	98,500
6120	120	195	89	27 x 120	113 x 126	85	100	117,500
780	134	156	99	32 x 80	120 x 87	95	60	106,000
7100	134	177	99	32 x 100	120 x 107	95	80	126,500
7120	134	197	99	32 x 120	120 x 127	95	100	150,000

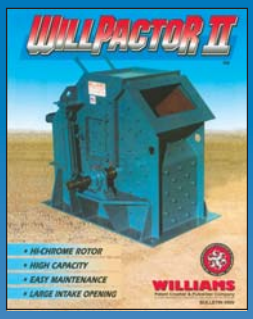
PERFORMANCE TABLE 3

SIZE	USUAL RANGE (RPM)	USUAL MOTOR SIZE (HP)	MAXIMUM FEED SIZE (inches)	USUAL RANGE GROSS CAPACITY (tons per hour)
111	1800 - 3600	5 -- 15	3	8 -- 18
215	1200 - 1800	20 -- 40	5	30 -- 50
220	1200 - 1800	25 -- 50	5	35 -- 65
230	1200 - 1800	40 -- 75	5	60 -- 95
240	1200 -1800	50 -- 100	5	75 -- 125
320	1100 - 1400	50 -- 125	8	65 -- 125
330	1100 - 1400	75 -- 200	8	100 -- 250
340	1100 - 1400	100 -- 250	8	125 -- 315
350	1100 - 1400	150 -- 300	8	190 -- 380
360	1100 - 1400	200 -- 400	8	250 -- 510
370	1100 - 1400	250 -- 450	8	320 -- 570
380	1100 - 1400	350 -- 500	8	450 -- 630
430	700 -- 900	100 -- 300	10	120 -- 370
445	700 -- 900	150 -- 400	10	190 -- 490
460	700 -- 900	200 -- 600	10	250 -- 615
475	700 -- 900	300 -- 700	10	370 -- 860
490	700 -- 900	400 -- 800	10	500 -- 980
540	600 -- 900	250 -- 500	12	300 -- 600
560	600 -- 900	300 -- 700	12	360 -- 850
580	600 -- 900	500 -- 1000	12	600 -- 1200
5100	600 -- 900	800 -- 1250	12	950 -- 1500
5120	600 -- 900	1000 - 1500	12	1200 -- 1800
640	500 -- 800	400 -- 600	15	480 -- 725
660	500 -- 800	500 -- 900	15	600 -- 1090
680	500 -- 800	700 -- 1250	15	850 -- 1500
6100	500 -- 800	900 -- 1500	15	1100 -- 1800
6120	500 -- 800	1250 - 1800	15	1500 -- 2170
780	450 -- 720	1000 - 500	20	1200 -- 1800
7100	450 -- 720	1500 - 2000	20	1800 -- 2400
7120	450 -- 720	1750 - 2250	20	2100 -- 2700

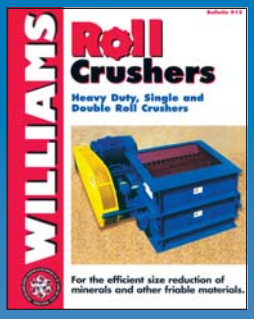
Call us to see our complete line of Williams Crushers & Equipment!



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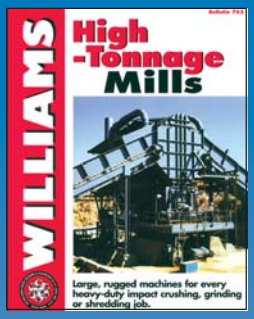
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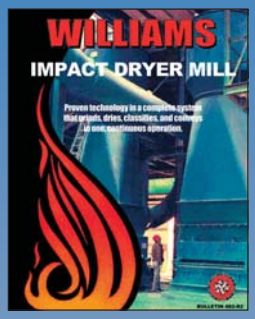
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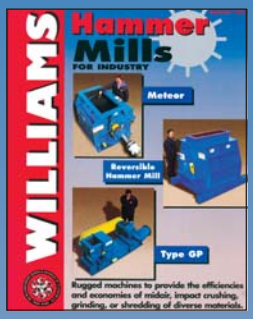
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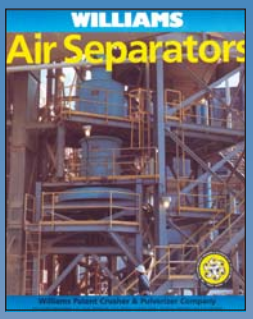
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BULLETIN 856R



BULLETIN 766R



BULLETIN 774

The Williams Way...

At Williams, the company's mission is to provide highly personalized engineering that starts with a complete understanding of the size reduction need and ends with a machine designed and built to meet it.

Along the way, a CAD/CAM interactive graphics computer system is emploted which permits three-dimensional, "on-screen" design of a part and translates it into numerical control tapes for operation of burning tables and machine tools.

Try the Williams Way. You'll like it. See your Williams representative or contact us directly.

Some of the materials being processed in Williams Reversible Impactors:

Anhydrous Boric Acid	Coal & Lignite	Limestone	Sand
Light Weight Aggregate	Coke	Lithium Ore	Shale
Aluminum Oxide	Coke, Petroleum Calcined	Magnesium Sludge	Silica
Barytes	Diatomacious Earth	Magnesite	Slags
Bauxite	Dolomite	Magnesite, Dead Burnt	Sodium Sulphate
Boric Acid	Ferro Silicon	Phosphorous Compounds	Trona Ore
Boron Ore	Fertilizer	Phosphate Rock	Tungsten Carbide
Brick (Cobbings)	Glass	Potassium Chloride	Tungsten, Sintered
Clay	Gravel	Potash & Potash Ore	
Cement Rock	Burnt Lime	Potash Flakes	

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